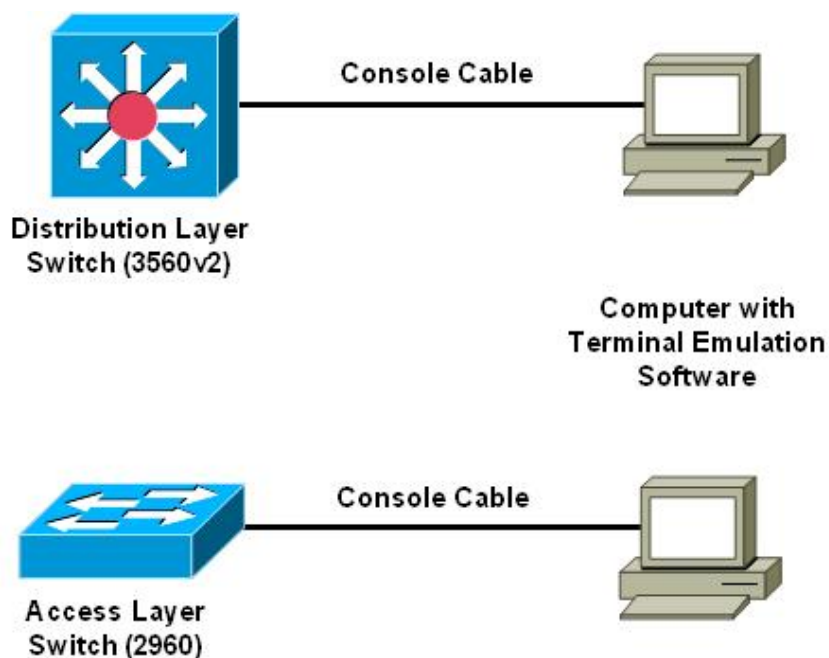


CCNPv7.1 SWITCH

Chapter 1 Lab - Preparing the Switch

Topology



Objectives

- Clear the configuration of **all the switches in your pod**
- Configure the database template used by **all the switches in your pod**
- Save a baseline configuration **for all the switches in your pod**

Background

When working with a switch that has been previously configured, any new commands entered will be merged with the existing configuration, causing unpredictable results. Additionally, if the switch is connected to other switches in the network, you can remove the VLANs but they might be relearned from another switch via VTP. In this lab you prepare your switches for use with future labs. This is accomplished by erasing the startup configuration from NVRAM and deleting the VLAN database. You also ensure that VLANs will not be relearned from another switch after the VLAN database has been deleted. Additionally, your switches may be required to support IPv6 traffic, which it does not by default. This is accomplished by changing the database template used by the Switch Database Manager.

Note: This lab was based on Cisco Catalyst 3560 and 2960 switches running Cisco IOS 15.0(2)SE6 IP Services and LAN Base images, respectively. **Catalyst 3650 switches (running any Cisco IOS XE release) and Catalyst 2960-Plus switches (running any comparable Cisco IOS image) are used in place of the Catalyst**

3560 switches and the Catalyst 2960 switches. They have SDM templates “advanced” and “lanbase-routing” respectively.

Required Resources

You may use one of the following switches or a comparable one with this lab:

- Cisco 2960 with the Cisco IOS Release 15.0(2)SE6 C2960-LANBASEK9-M or comparable
- Cisco 3650 with the Cisco IOS XE Release CAT3k_CAA-UNIVERSALK9 with ipservicesk9 package installed or comparable
- Lab Computer with PuTTY software

Step 1: Connect to the switch console port and enter privileged EXEC mode.

From a computer running a terminal emulation program, connect to the console port of the switch that you want to clear using a console cable. You should see a console prompt that includes the switch’s hostname, followed by a > or #. The default switch hostname is “Switch.”

```
Switch>
```

or

```
Switch#
```

If the prompt ends with a >, you are not in privileged EXEC mode. To enter privileged EXEC mode, type **enable**. This might require a password. If you are in a configuration mode, type **exit** or **end**.

If not enabled:

```
Switch> enable
```

```
Switch#
```

If in global configuration mode:

```
Switch(config)# exit
```

```
Switch#
```

Step 2: Delete the VLAN database file, if present.

A VLAN database file named vlan.dat might exist in FLASH on the switch if it has been previously used in the network. This file holds information about VLANs created on the switch, their IDs, names, types and states, and it also stores the VTP settings. In privileged EXEC mode, type **dir** or **dir flash:** and press Enter. This will provide a directory listing of the files in FLASH. In particular, note two files in the output: the vlan.dat file that will be removed in this step, and the multiple-fs file that will be explained and removed in Step 3 below.

2960

```
Switch# dir
```

```
Directory of flash:/
```

2	-rw-	14641848	Mar 1 1993 00:05:34 +00:00	c2960-lanbasek9-mz.152-2.E7.bin
3	-rw-	34	Mar 1 1993 00:40:30 +00:00	pnp-tech-time
4	-rw-	34477	Mar 1 1993 00:40:40 +00:00	pnp-tech-discovery-summary
5	-rw-	2072	Mar 1 1993 00:01:06 +00:00	multiple-fs
6	-rw-	616	Mar 1 1993 01:13:09 +00:00	vlan.dat

3650Switch# **dir**

Directory of flash:/

```
46472 -rw-          409853870 Feb 13 2018 00:11:29 +00:00 cat3k_caa-
universalk9.16.06.02.SPA.bin
54209 drwx           4096 Jan 14 2015 03:21:46 +00:00 .installer
46466 -rw-           370 Jan 14 2015 03:22:00 +00:00 bootloader_evt_handle.log
61953 drwx           4096 Nov 1 2014 01:37:08 +00:00 core
69698 drwx           4096 Nov 1 2014 01:33:33 +00:00 .prst_sync
7745 drwx            4096 Nov 1 2014 01:33:37 +00:00 .rollback_timer
15489 drwx           4096 Nov 1 2014 01:33:45 +00:00 gs_script
23233 drwx           4096 Jan 14 2015 03:22:17 +00:00 dc_profile_dir
46467 -rw-          98869 Jan 14 2015 03:22:38 +00:00 memleak.tcl
46468 -rw-          2097152 Jan 14 2015 04:11:29 +00:00 nvram_config
46469 -rw-          2097152 Jan 14 2015 04:11:30 +00:00 nvram_config_bkup
46470 -rw-           35 Jan 14 2015 04:01:24 +00:00 pnp-tech-time
46471 -rw-          124337 Jan 14 2015 04:01:27 +00:00 pnp-tech-discovery-summary
46473 -rw-           616 Jan 14 2015 04:11:21 +00:00 vlan.dat
```

In privileged EXEC mode, type **delete flash:vlan.dat** or **delete vlan.dat** (in the shorter form without the `flash:` prefix, the <TAB> key completion does not work) and press Enter. If you are asked to confirm, press Enter until you are back to the original prompt.

Switch# **delete vlan.dat**

Delete filename [vlan.dat]? ! Press Enter

Delete flash:vlan.dat? [confirm] ! Press Enter

Switch#

Step 3: Erase the startup config from NVRAM.

On Cisco devices, NVRAM is the common location for configuration files. The goal of this step is to entirely erase the NVRAM contents so that on the next boot, the switch starts in a factory default configuration. There are, however, a few important facts you need to be aware of.

On the switch platforms used in these or similar labs, such as Catalyst 2950, 2960, 3550, 3560, 3650, 3750, 3850, the NVRAM is not truly physically present. Instead, a part of the FLASH memory is used to store the NVRAM contents. In other words, on these switch platforms, the NVRAM is only simulated using a part of the FLASH, as also evidenced by one of lines in the **show version** command output:

2960Switch# **show version | include volatile**

64K bytes of flash-simulated non-volatile configuration memory.

3650Switch# **show version | include volatile**

2048K bytes of non-volatile configuration memory.

Note: The XE images of the 3650 actually has an NVRAM directory in Flash.

Files that appear to reside in NVRAM (use **dir nvram:** to display its contents) are in fact stored in FLASH. Some of them are stored as standalone files in FLASH, such as flash:config.text that maps to nvram:startup-config, or flash:private-config.text that maps to nvram:private-config and stores sensitive information such as RSA keys, master password encryption key etc. Deleting any of these files from FLASH will cause the corresponding mapped file in simulated NVRAM to also be deleted or its apparent length in NVRAM to be truncated to zero, and vice versa. Other files in NVRAM, such as self-generated X.509 certificates, are all stored in the flash:multiple-fs file.

The **erase startup-config** command commonly used to remove the stored configuration will remove the flash:config.text and flash:private-config.text. However, other contents of the simulated NVRAM, such as X.509 certificates that were automatically created for the HTTPS server run on the switch, will not be removed as they reside in the flash:multiple-fs file unaffected by the **erase startup-config** command. Therefore, to completely erase the simulated NVRAM contents, not only the **erase startup-config** command must be issued, but also the flash:multiple-fs file must be removed.

In privileged EXEC mode, issue the **delete flash:multiple-fs** (or simply **delete multiple-fs**) command, followed by the **erase startup-config** command. Press Enter on each prompt.

```
Switch# delete multiple-fs
Delete filename [multiple-fs]?                ! Press Enter
Delete flash:/multiple-fs? [confirm] ! Press Enter
Switch# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
                                           ! Press Enter

[OK]
Erase of nvram: complete
Switch#
*Mar  1 00:43:23.286: %SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
```

Note: Historically, the **write erase** command was used to erase the startup configuration. This command has the same effect as **erase startup-config** command that was implemented in later IOS versions, and is still being used as its shortened version **wr e** is more convenient to write than **erase startup-config**.

Step 4: Change the Switch Database Template.

The Cisco Switch Database Manager (SDM) provides various TCAM allocation templates that can be enabled to support specific roles, depending on how the switch is used in the network. By default **3650** switch is using the “Advanced ” template. This particular template divides the available TCAM up for use by the different processes and protocols in a manner that most likely supports standard IPv4 and IPv6 unicast and multicast traffic. The older 3560 and the 2960 switches used the “default” template which only supports standard IPv4 unicast and multicast traffic. Use the **show sdm prefer** command to see the details of the current template. Output may differ depending on the specific switch platform.

NOTE: the output below refers to the 3560 switch. The 3560 switches are in ATC330. It has been left in for historical purposes, so you know how other switches may work. The 3650 output will be shown after.

3560 Switch

```
Switch# show sdm prefer
The current template is "desktop default" template.
The selected template optimizes the resources in
the switch to support this level of features for
```

8 routed interfaces and 1024 VLANs.

number of unicast mac addresses:	6K
number of IPv4 IGMP groups + multicast routes:	1K
number of IPv4 unicast routes:	8K
number of directly-connected IPv4 hosts:	6K
number of indirect IPv4 routes:	2K
number of IPv6 multicast groups:	0
number of IPv6 unicast routes:	0
number of directly-connected IPv6 addresses:	0
number of indirect IPv6 unicast routes:	0
number of IPv4 policy based routing aces:	0
number of IPv4/MAC qos aces:	0.5K
number of IPv4/MAC security aces:	1K
number of IPv6 policy based routing aces:	0
number of IPv6 qos aces:	20
number of IPv6 security aces:	25

Switch#

Notice in the output that there is NO memory allocated to IPv6 operations.

3650 Switches - These switches are used in ATC328/9

Switch#sh sdm prefer

Showing SDM Template Info

This is the Advanced template.

Number of VLANs:	4094
Unicast MAC addresses:	32768
Overflow Unicast MAC addresses:	512
IGMP and Multicast groups:	4096
Overflow IGMP and Multicast groups:	512
Directly connected routes:	16384
Indirect routes:	7168
Security Access Control Entries:	3072
QoS Access Control Entries:	2560
Policy Based Routing ACEs:	1024
Netflow ACEs:	768
Wireless Input Microflow policer ACEs:	256
Wireless Output Microflow policer ACEs:	256
Flow SPAN ACEs:	256
Tunnels:	256

Control Plane Entries:	512
Input Netflow flows:	8192
Output Netflow flows:	16384
SGT/DGT and MPLS VPN entries:	3840
SGT/DGT and MPLS VPN Overflow entries:	512

These numbers are typical for L2 and IPv4 features.

Some features such as IPv6, use up double the entry size;
so only half as many entries can be created.

There are several different SDM templates available for use, each with different amounts of TCAM allocated to different processes and protocols. Use the **show sdm prefer *template*** command to examine the details of a particular database template.

3560 Switch

```
Switch# show sdm prefer ?
```

```
access          Access bias
default         Default bias
dual-ipv4-and-ipv6  Support both IPv4 and IPv6
routing         Unicast bias
vlan           VLAN bias
|              Output modifiers
<cr>
```

Because our switching network will need to support IPv6 traffic, the SDM template must be changed to one of the dual-ipv4-and-ipv6 templates for the 3560 switch, using the **sdm prefer dual-ipv4-and-ipv6 *template*** global configuration command. **For the 3650 switch we don't need to change the SDM template, as it already supports IPv4 and IPv6.**

The template options for dual IPv4 and IPv6 operation vary based on the model of switch. As of this writing, the dual-ipv6-and-ipv4 template options on 3560s are **default**, **routing**, and **VLAN**, while the option on 2960s is **default**. The 2960's **lanbase-routing** template also supports connected IPv6 hosts.

Note: The **routing** template is not a valid selection on switches running the LANBASE feature set, even though **routing** may appear as an option at the command line.

2960 Switch only

```
Switch(config)# sdm prefer ?
```

```
default         Default bias
dual-ipv4-and-ipv6  Support both IPv4 and IPv6
lanbase-routing  Supports both IPv4 and IPv6 Static Routing
qos             QoS bias
```

```
Switch(config)# sdm prefer lanbase-routing
```

Changes to the running SDM preferences have been stored, but cannot take effect until the next reload.

Use 'show sdm prefer' to see what SDM preference is currently active.

As you can see in the output above, a restart is required for the change to be effective; the switch cannot re-allocate the TCAM on the fly. You will restart the switch at the end of this lab.

Configure your 2960 switches for the **lanbase-routing** template. **For the 3650 switch, we do not need to change the SDM template.**

Note: Several of the labs in the course end with instructions to reset the switch to its defaults. If you clear the switch (**delete vlan.dat, delete multiple-fs, write erase, reload**), the selected SDM template will return to the default, and could require reconfiguration (including a reboot).

Step 5: Reload the device, but do not save the system configuration if prompted.

After clearing the switch configuration, reload the switch by typing **reload** and pressing Enter. If you are asked whether to save the current configuration, answer **no**. Press Enter to confirm. The switch starts reloading. Your output might look different depending on the switch model that you are using. This step might take a few minutes, because the switch needs time to reload.

```
Switch# reload
```

```
System configuration has been modified. Save? [yes/no]: no
Proceed with reload? [confirm]
```

```
*Mar 11 23:03:06.985: %SYS-5-RELOAD: Reload requested by console. Reload Reason:
Reload command.
<output omitted>
```

Step 6: Create a Baseline Configuration

To eliminate some of the redundant basic configurations, use a TCL script to build a configuration shell. This can be customized several ways, but the basic text below creates a file named BASE.CFG in FLASH that can be used at the beginning of labs after the switch has been completely cleared. This will save some of the mundane configuration steps

Use the script below and modify it to meet the particulars of the switch you are working on:

ALS1 BASE.CFG

```
tclsh
puts [ open "flash:BASE.CFG " w+ ] {
hostname ALS1
ip domain-name CCNP.NET
no ip domain-lookup
```

```
interface range fa0/1-24 , g0/1-2
shutdown
exit
vtp mode transparent
line con 0
no exec-timeout
logging synchronous
exit
end
}
tclquit
```

ALS2 BASE.CFG

```
tclsh
puts [ open "flash:BASE.CFG " w+ ] {
hostname ALS2
ip domain-name CCNP.NET
no ip domain-lookup
interface range fa0/1-24 , g0/1-2
shutdown
exit
vtp mode transparent
line con 0
no exec-timeout
logging synchronous
exit
end
}
tclquit
```

DLS1 BASE.CFG

```
tclsh
puts [ open "flash:BASE.CFG" w+ ] {
hostname DLS1
ip domain-name CCNP.NET
no ip domain-lookup
interface range g1/0/1-24 , g1/1/1-4
```

```
shutdown
exit
vtp mode transparent
line con 0
no exec-timeout
logging synchronous
exit
end
}
tclquit
```

DLS2 BASE.CFG

```
tclsh
puts [ open "flash:BASE.CFG" w+ ] {
hostname DLS2
ip domain-name CCNP.NET
no ip domain-lookup
interface range g1/0/1-24 , g1/1/1-4
shutdown
exit
vtp mode transparent
line con 0
no exec-timeout
logging synchronous
exit
end
}
tclquit
```

Step 7: Create a script to automate clearing and reloading of the switch

Use TCL once more to create a script to automate the tasks involved in clearing the switch.

DLS1/DLS2:

```
tclsh
puts [ open "flash:dls_reset.tcl" w+ ] {
typeahead "\n"
copy running-config startup-config
typeahead "\n"
erase startup-config
delete /force vlan.dat
delete /force multiple-fs
typeahead "\n"
puts "Reloading the switch in 1 minute, type reload cancel to halt"
typeahead "\n"
reload in 1
}
tclquit
```

run this script by entering "tclsh dls_reset.tcl"

ALS1/ALS2:

```
tclsh
puts [ open "flash:als_reset.tcl" w+ ] {
typeahead "\n"
copy running-config startup-config
typeahead "\n"
erase startup-config
delete /force vlan.dat
delete /force multiple-fs
ios_config "sdm prefer lanbase-routing"
typeahead "\n"
puts "Reloading the switch in 1 minute, type reload cancel to halt"
typeahead "\n"
reload in 1 ALS_RESET.TCL SCRIPT RUN
}
Tclquit
```

run this script by entering "tclsh als_reset.tcl"

Now for any lab that requires clearing the switch, issue the privileged exec command **tclsh dls_reset.tcl** or **als_reset.tcl** and the switch will be completely cleared and reload in the proper state.

Step 8: End of Lab

At this point, your switches should be at a factory default with the BASE.CFG file in FLASH for future use.